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申请人姓名：

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论文作者签名： ______________________

指导教师签名： ______________________

论文评阅人 1： ______________________

评阅人 2： ______________________

评阅人 3： ______________________

评阅人 4： ______________________

评阅人 5： ______________________

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Abstract

Modern diffusion-based video super-resolution (VSR) can process footage that runs into minutes, but the field's evaluation instruments cannot: full-reference metrics require ground truth that deployment videos lack, adjacent-frame temporal metrics reward spatial smoothness rather than consistency, and clip-averaged perceptual benchmarks discard the cross-clip drift signal that long videos accumulate. This thesis characterises these failures empirically and builds the missing instrument.

First, a systematic failure-mode study on a numerically validated evaluation environment establishes four structural breakdowns: a content-dependent regime shift in which two state-of-the-art learned metrics invert method rankings against human judgement on close-up content; a temporal-scale crossover in which the conventional tOF metric's winner reverses between adjacent-frame and long-range lags; a categorical blind spot in which slow colour drift is invisible to eight widely-used metrics; and a smoother-output bias so strong that a state-of-the-art consistency benchmark ranks a degraded input above its own super-resolutions.

Second, the thesis proposes LR-VCC (Long-Range Video Consistency Composite), a no-reference metric composing seven sub-metrics — appearance, multi-scale temporal warping, identity, pairwise and anchored colour stability, semantic trajectory drift, and colour-trajectory slope — through a reliability-weighted softmax-log-mean, in which every sub-metric is gated by an interpretable regime indicator. Validated on a twelve-family synthetic artefact benchmark with designed positive and null controls, LR-VCC closes the colour-drift blind spot, fixes a newly characterised convergence-rewarding pathology of stability metrics via anchored measurement, and preserves correct method rankings where existing metrics flip. Applied to real methods (MGLD-VSR, Upscale-A-Video, FlashVSR), it yields a stable ranking with an actionable signature: the streaming method wins identity preservation while exhibiting the worst long-range semantic drift. Hyperparameter-sensitivity and leave-one-out studies quantify the robustness of each claim.

Third, a causal study localises that drift. A bit-exact position-injection instrument decomposes rotary position embedding (RoPE) extrapolation into absolute magnitude and relative geometry, per axis, scored against real ground truth. Absolute position offsets are quality-free even fifty times beyond the trained window; relative-distance dilation degrades quality gracefully in time but steeply in space; real resolution extrapolation is free to twice the trained extent; and the model's long-video drift is demonstrably not positional — exonerating position encoding and redirecting improvement effort to streaming generation mechanics. The study also demonstrates that self-consistency does not predict quality, and provides an empirical, per-axis account of when position interpolation helps and when it hurts.

Keywords: video super-resolution; long video; temporal consistency; no-reference video quality assessment; rotary position embedding; diffusion models

摘要

现代基于扩散模型的视频超分辨率（VSR）已能处理长达数分钟的视频，但现有评价体系尚无法与之匹配：全参考指标依赖部署场景中并不存在的高分辨率真值；相邻帧时序指标实际奖励的是空间平滑度而非时序一致性；按片段平均的感知基准则抹去了长视频逐渐累积的跨片段漂移信号。本文对上述失效进行了系统的实证刻画，并构建了缺失的评价工具。

首先，本文在数值上经过严格校验的评价环境中确立了四类结构性失效：两个最先进的学习型指标在特写内容上出现与人类判断相反的排序反转（内容依赖的工作域漂移）；常用时序指标 tOF 的优胜方法在相邻帧与长程时距之间发生反转（时间尺度交叉）；缓慢颜色漂移对八种常用指标完全不可见（类别性盲区）；以及”平滑输出偏置”——其影响之强，使得最先进的一致性基准将退化输入排在其超分结果之上。

其次，本文提出 LR-VCC（长程视频一致性复合指标）：一种无参考指标，通过可靠性加权的 softmax 对数均值组合七个子指标——外观质量、多尺度时序变形误差、人物身份、成对与锚定颜色稳定性、语义轨迹漂移及颜色轨迹斜率——其中每个子指标均由可解释的工作域指示量进行可靠性门控。在包含设计性阳性与阴性对照的十二类合成伪影基准上，LR-VCC 消除了颜色漂移盲区，通过锚定测量修复了本文新刻画的”奖励收敛”病理，并在现有指标发生反转的内容上保持了正确的方法排序。将其应用于真实方法（MGLD-VSR、Upscale-A-Video、FlashVSR）得到了稳定的排序及可操作的特征：流式方法在身份保持上最优，却在长程语义漂移上最差。超参数敏感性与留一消融实验量化了各项结论的稳健性。

最后，本文以因果实验定位了上述漂移。通过逐位精确的位置注入工具，将旋转位置编码（RoPE）的外推分解为绝对位置幅度与相对距离几何两个因素，逐轴以真实真值评分。结果表明：绝对位置偏移即使超出训练窗口五十倍也不损失质量；相对距离的拉伸在时间轴上退化平缓、在空间轴上退化陡峭；真实分辨率外推至训练范围两倍仍无质量损失；且该模型的长视频漂移可证明与位置编码无关——从而为位置编码”平反”，并将改进方向指向流式生成机制。研究还证明自一致性无法预测质量，并给出了位置插值何时有效、何时有害的逐轴实证刻画。

关键词: 视频超分辨率; 长视频; 时序一致性; 无参考视频质量评价; 旋转位置编码; 扩散模型

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Chapter 1. Introduction and Background

1.1 Introduction

Video super-resolution (VSR) has advanced rapidly with the adoption of diffusion-based and transformer-based architectures. Where earlier CNN-based methods upscaled fixed short clips (typically 7–32 frames), modern methods such as MGLD-VSR[1], Upscale-A-Video[2], and the one-step streaming design FlashVSR[3] can process substantially longer sequences in latent or sliding-window fashion, enabling deployment on real-world footage that runs into minutes rather than seconds.

This shift toward long-video SR introduces an evaluation gap. The standard VSR benchmarks — REDS[4] (100-frame clips at 720×1280), Vimeo-90K[5] (7-frame septuplets), Vid4[6], UDM10[7], and SPMCS[8] — were designed for short-clip evaluation and report full-reference metrics (PSNR, SSIM[9], LPIPS[10]) against paired HR ground truth. When applied to long videos in the wild, three fundamental limitations emerge:

- **No HR ground truth.** Real deployment videos have no paired high-resolution reference, making full-reference metrics (PSNR, SSIM, LPIPS, DISTS[11]) inapplicable.
- **Adjacent-frame temporal metrics miss long-range artefacts.** Existing temporal metrics — tOF and tLP from TecoGAN[12], and E^*warp from DOVE[13] — are computed at a single adjacent-frame gap (k=1). A method that is locally smooth but drifts globally over minutes is undetectable by these measures.
- **Smoother-output bias in learned representations.** DINOv2-based[14] consistency metrics and LPIPS favour methods whose outputs change less in feature space between frames. Diffusion-based methods produce detail-rich outputs that score worse on these metrics despite being visually superior, a structural bias that misrepresents quality for modern SR architectures.

More precisely, the evaluation of long-form video super-resolution is structurally insufficient in four overlapping ways, documented empirically in Chapter 3. First, per-frame fidelity metrics (PSNR, SSIM, LPIPS, DISTS, CLIP-IQA) are temporal-blind by construction: a video that drifts slowly in colour across thousands of frames, or that jumps at every tile boundary of a tile-based diffusion super-resolver, or whose face identity degrades from minute one to minute two, scores identically to a colour-stable, seam-free, identity-preserving video provided each individual frame is locally crisp. Second, short-range temporal metrics (tOF, tLP, E^*warp at adjacent-frame lag $k=1$) measure adjacent-frame smoothness but cannot see consistency across the longer time horizons where the most damaging long-video failures actually occur, and they systematically favour spatially smoother output. Third, perceptual aggregates (DOVER[15], VBench-2.0[16]) average clip-level scores up to a per-video number, removing the very cross-clip consistency signal that long-video evaluation requires. Fourth, even metrics that should catch a failure mode exhibit content-dependent calibration shifts that are not surfaced to the user: the same metric, under identical hyperparameters, can flip its method ranking on particular content (close-up faces, dense motion) because a learned classifier is pushed into a different operating regime.

A researcher who wishes to know whether method A is more temporally consistent than method B on long-form content has, today, no single trustworthy answer. This thesis addresses that gap.

1.2 Research Aim, Hypotheses, and Contributions

The primary aim of this research is to develop and validate a no-reference long-range video consistency metric for long-form video super-resolution. The metric must (i) catch the long-range failure modes to which existing metrics are demonstrably blind; (ii) preserve correct per-video method rankings on real super-resolved outputs, including on content where existing learned metrics flip against human judgement; and (iii) be reproducible, modular, and extensible. The metric introduced for this purpose is the **Long-Range Video Consistency Composite (LR-VCC)**, specified in Chapter 4.

The metric research is organised around three falsifiable hypotheses:

H1 (Composition). Reliability-weighted softmax-log-mean composition of independent sub-metrics catches failure modes that no single constituent sub-metric catches.

H2 (Multi-scale temporal). Aggregating temporal warping error over multiple frame gaps catches both high-frequency and long-range temporal failure modes that fixed-scale temporal metrics miss.

H3 (Reliability gating). Per-sub-metric reliability gates derived from interpretable signals reduce content-dependent ranking flips of the type observed in the close-up regime-shift case study.

A second research question arose from applying LR-VCC to real models. The streaming one-step diffusion model FlashVSR ranks within 0.012 of the top method overall and wins the identity sub-metric outright — yet simultaneously exhibits the worst long-range semantic-drift sub-metric score of the three methods evaluated, precisely the failure signature the metric was designed to expose. Because FlashVSR's streaming design lets absolute rotary position embedding (RoPE) indices grow without bound as video length increases, position encoding is the natural suspect:

RQ (Positional extrapolation). How does RoPE extrapolate beyond its trained window in time and in space in a streaming diffusion VSR model, and is positional extrapolation the cause of the observed long-video and high-resolution quality degradation?

Chapter 6 answers this question causally, using a bit-exact position-injection instrument that manipulates position indices independently of video content.

This thesis makes five contributions:

1. **A systematic failure-mode characterisation of existing long-video SR metrics** (Chapter 3): a content-dependent regime shift in which two VBench-2.0 metrics flip rankings against visual judgement on close-up content; a temporal-scale crossover in which the tOF winner inverts between $k{=}1$ and $k{\geq}10$, demonstrating that adjacent-frame temporal metrics measure spatial smoothness rather than consistency; and a severity-response

study showing that of 32 (metric, artefact family) cells across eight widely-used metrics, only five respond cleanly — with slow colour drift invisible to every baseline metric tested.

2. **LR-VCC, a no-reference reliability-gated composite metric** (Chapter 4): seven sub-metrics covering appearance, multi-scale temporal consistency, identity preservation, colour-distribution stability, anchored colour stability, semantic trajectory drift, and colour-trajectory slope, composed by a reliability-weighted softmax-log-mean. Every sub-metric is gated by an interpretable reliability signal, and every design iteration closed a specific characterised failure mode.

3. **A parameterised synthetic consistency benchmark for long-video SR** (Chapter 5): twelve artefact families (drift, boundary, flicker, identity, background, and flip/control families) at five severities over five long base videos, with generators verified for zero-severity identity and seed reproducibility.

4. **A multi-method real-model evaluation** (Chapter 5): LR-VCC applied to MGLD-VSR, Upscale-A-Video, and FlashVSR on long real-SR outputs, yielding a ranking (MGLD-VSR $0.622 >$ FlashVSR $0.610 >$ UAV 0.589) that per-frame metrics do not produce, and exposing FlashVSR's distinctive signature: best identity preservation, worst long-range semantic drift. Hyperparameter-sensitivity and leave-one-out studies quantify which parts of the ranking are robust and which sub-metric drives each verdict.

5. **A causal analysis of RoPE extrapolation in streaming diffusion VSR** (Chapter 6): absolute-position offsets are quality-free even $50\times$ beyond the trained window on every axis; relative-distance geometry dilation damages quality monotonically while mild compression is free; the model's long-video drift is demonstrably <u>not</u> positional; and real spatial grid extension to $1.5\times$ costs only -0.15 dB while spatial position interpolation there <u>hurts</u>. These verdicts exonerate position encoding and redirect improvement effort toward streaming cache/generation mechanics.

1.3 Related Work

1.3.1 Video Super-Resolution Methods

Modern VSR methods fall into three architectural families. Recurrent CNN methods such as BasicVSR++[17] propagate features bidirectionally with flow-guided alignment; RealBasicVSR[18] extends the family to real-world degradations and contributes the degradation pipeline used by later benchmarks. Transformer methods — VRT[19] and its recurrent variant RVRT[20] — model temporal dependencies with attention but face quadratic cost in sequence length, restricting them to short clips. Diffusion-based methods leverage generative priors for perceptual quality: StableSR[21] adapts a text-to-image diffusion prior to image SR; Upscale-A-Video[2] adds temporal layers for video; MGLD-VSR[1] guides latent diffusion with optical-flow motion cues; and FlashVSR[3] distils a video diffusion transformer[22] into a one-step streaming super-resolver capable of arbitrary-length input. The two primary baselines evaluated throughout this thesis, MGLD-VSR and Upscale-A-Video, represent the detail-preserving and smoothing ends of the diffusion VSR spectrum respectively; FlashVSR represents the emerging streaming class. Their architectures are detailed in Chapter 2.

1.3.2 Video Quality Assessment

The existing metric ecosystem can be organised into five families:

1. **Full-reference pixel/perceptual** (PSNR, SSIM[9], LPIPS[10], DISTS[11]) — require HR ground truth; inapplicable to deployment-time long-video evaluation.

2. **No-reference single-aspect** (CLIP-IQA[23], MUSIQ[24], NIQE, BRISQUE) — measure per-frame quality; no temporal dimension.

3. **Temporal pixel-level** (tOF, tLP from TecoGAN[12]; E^*warp from DOVE[13]) — adjacent-frame (k=1) optical-flow consistency; systematically miss long-range temporal structure.

4. **Semantic video quality** (VBench[25], VBench-2.0[16]) — multi-aspect benchmarks covering subject consistency, Human_Identity, Human_Anatomy, and others. These bench-

marks were designed for text-to-video generation evaluation; thirteen of VBench-2.0's eighteen dimensions require text prompts and are therefore inapplicable to super-resolution, while the remaining prompt-independent dimensions operate at clip level and average per-clip scores up to a single video number, removing the cross-clip drift signal that long-video consistency assessment requires. Content-dependent failure modes are documented in Chapter 3.

5. **Disentangled aesthetic-technical no-reference** (DOVER[15]) — a learned video quality predictor combining an aesthetic branch and a technical branch. DOVER is per-video and produces a single aggregate; it does not specifically model cross-frame temporal drift.

No existing metric satisfies all four desiderata simultaneously: (1) no-reference, (2) multi-time-scale temporal coverage, (3) robust to smoother-output bias from learned representations, and (4) reliable across diverse long-video content including close-up scenes. LR-VCC (Chapter 4) is designed to satisfy all four.

1.3.3 Position Encodings and Length Extrapolation

Rotary position embeddings (RoPE)[26] encode token positions as rotations applied to query and key vectors, making attention scores a function of relative position. RoPE underlies most modern large language models and, in a three-axis factorised form (time, height, width), most video diffusion transformers including Wan2.1[22], the backbone of FlashVSR. A model trained on a fixed context window degrades when evaluated beyond it; the language-modelling community has produced a family of position interpolation remedies — linear position interpolation (PI)[27], NTK-aware frequency scaling[28], and YaRN[29] — that compress or re-scale position indices so that relative distances remain inside the trained range. Whether these findings transfer to video diffusion transformers, where positions are three-dimensional and streaming inference lets the temporal index grow without bound, has not been systematically studied. Chapter 6 provides such a study for a production streaming VSR model, decomposing extrapolation into absolute-position magnitude and relative-distance geometry and measuring each against real ground truth.

1.3.4 Long-Video Modelling and Evaluation Infrastructure

Long-horizon video modelling is an active frontier: state-space world models[30] extend temporal memory to thousands of frames with linear-cost scanning, and streaming diffusion designs such as FlashVSR bring arbitrary-length processing to restoration. Evaluation infrastructure has not kept pace: the DOVE suite[13] standardises realistic degradation and full-reference scoring but on short clips, and VBench's long-video mode re-aggregates clip-level scores, discarding cross-clip drift. The gap between long-video generation capability and long-video evaluation capability is precisely where this thesis operates: LR-VCC supplies the missing multi-scale consistency instrument, and the RoPE study supplies the first per-axis positional-extrapolation characterisation of a production streaming VSR model.

1.4 Thesis Organisation

Chapter 2 describes the foundational technologies this thesis builds on: the evaluated VSR models, rotary position embeddings, and the perceptual and temporal measurement toolkit. Chapter 3 characterises the failure modes of existing metrics on long-video SR content. Chapter 4 specifies the LR-VCC metric and its design rationale. Chapter 5 validates LR-VCC on the synthetic benchmark and on three real SR methods. Chapter 6 presents the RoPE extrapolation study. Chapter 7 discusses significance and limitations, and concludes.

Chapter 2. Foundational Technologies

This chapter describes the models, position-encoding machinery, and measurement toolkit that the rest of the thesis builds on: the three evaluated diffusion VSR methods (Section 2.1), rotary position embeddings and their extrapolation behaviour (Section 2.2), the perceptual and temporal quality measures from which LR-VCC's sub-metrics are constructed (Section 2.3), and the DOVE evaluation protocol under which all ground-truth comparisons are performed (Section 2.4).

2.1 Diffusion-Based Video Super-Resolution and the Evaluated Baselines

Diffusion-based restoration adapts a generative prior — typically a text-to-image or text-to-video latent diffusion model — to the inverse problem of super-resolution: the degraded input conditions a denoising process that runs in the compressed latent space of a variational autoencoder, and the decoded result inherits the prior's ability to synthesise plausible high-frequency detail that no per-pixel regression objective would dare commit to[21]. The price is a new axis of failure: the synthesised detail must stay consistent — across tiles within a frame (large frames are processed in overlapping tiles), across frames within a clip, and across the minutes of a long video. The three evaluated methods make three different trades along this axis.

2.1.1 MGLD-VSR

MGLD-VSR[1] (ECCV 2024) applies motion-guided latent diffusion to progressively refine super-resolved frames, using optical-flow cues to steer the latent denoising trajectory toward temporal coherence. It is characterised by sharp, detail-preserving output. In this thesis it also serves as the numerically validated anchor of the evaluation environment: it reproduces the published DOVE benchmark numbers on UDM10 exactly (PSNR 24.23, with SSIM and LPIPS matching to four decimal places).

2.1.2 Upscale-A-Video

Upscale-A-Video (UAV)[2] (CVPR 2024) is a text-conditioned temporal diffusion model inspired by text-to-video generation. It produces smoother, perceptually cleaner output at the cost of some high-frequency detail. UAV reproduces DOVE UDM10 results within a consistent $+1.33$ dB PSNR offset attributable to a degradation-pipeline configuration difference rather than a setup defect; the offset is consistent across test sets ($+1.68$ dB on SPMCS), confirming the evaluation environment.

2.1.3 FlashVSR

FlashVSR[3] is a one-step streaming VSR model distilled from the Wan2.1 video diffusion transformer[22]. A causal video VAE compresses time $4\times$ (and space $16\times$ per axis after patchify), so 81 pixel frames become 21 latent frames — the DiT's trained temporal window. Inference streams the video in chunks: the first chunk processes six latent frames, and every subsequent chunk processes two, attending over a short KV cache (two current latents plus three cached windows, $\sim$8 latents $\approx$ 30 pixel frames of effective span) with block-sparse attention whose sparsity ratio adapts to resolution; a lightweight decoder reconstructs pixels in one denoising step. Crucially for this thesis, its position encoding is a three-axis factorised RoPE with per-axis precomputed frequency tables of 1,024 rows, and **the temporal RoPE index is the absolute latent index of the stream** — it grows monotonically and never resets, while the short cache keeps relative distances small. Absolute positions therefore extrapolate without bound on long videos (the stock table caps a single pass at 4,089 pixel frames), the trained spatial extent is 48 latent rows (from the 768×1408 distillation resolution), and the model becomes a natural laboratory for separating absolute-magnitude from relative-geometry extrapolation — the decomposition Chapter 6 exploits.

2.2 Rotary Position Embeddings and Length Extrapolation

Rotary position embeddings[26] encode a token's position m by rotating each two-dimensional channel pair of its query and key vectors by an angle $m\theta_i$, where θ_i is a per-pair frequency drawn from a geometric schedule. The attention logit between positions m and n then depends on positions only through the relative offset $(m - n)$, giving translation invariance in expectation while retaining absolute-position information within each head.

Video diffusion transformers factorise RoPE over three axes: the channel dimension is partitioned into a temporal group and two spatial groups (height, width), each with its own position index and frequency table. In Wan2.1-derived models such as FlashVSR the tables are precomputed to a fixed number of rows; positions beyond the table are unreachable at inference time unless the table is extended.

When a model is evaluated at positions or relative distances outside its trained range, quality can degrade. The language-modelling literature offers three main remedies: linear position interpolation (PI)[27] compresses positions by a constant factor so relative distances remain in-range; NTK-aware scaling[28] rescales the frequency schedule non-uniformly, preserving high-frequency (local) resolution; and YaRN[29] combines frequency-dependent interpolation with attention-temperature correction. All three were developed for one-dimensional text and validated on perplexity; their applicability to three-axis video RoPE under streaming inference is one of the questions Chapter 6 answers empirically.

2.3 Perceptual and Temporal Quality Measurement Toolkit

Three families of measures used throughout the thesis are described here.

No-reference perceptual measures. CLIP-IQA[23] evaluates per-frame image quality using CLIP[31] embeddings. Unlike LPIPS or DINOv2, CLIP-IQA does not systematically penalise high-frequency detail in diffusion-style SR outputs, making it a suitable appearance component for a no-reference composite. MUSIQ[24] is a multi-scale image-quality transformer; DOVER[15] predicts video quality through disentangled aesthetic and technical branches.

Temporal consistency measures. tOF and tLP[12] measure optical-flow warping error and LPIPS-based perceptual warping error between frame pairs. Conventionally both are evaluated at adjacent-frame lag k=1; this thesis extends them to lags $k \in \{1, 5, 10, 30, 60, 120\}$ using

RAFT[32] optical flow with forward–backward consistency masking applied uniformly across all k.

Identity measures. VBench-2.0's Human_Identity dimension tracks face embedding stability using RetinaFace[33] detection and ArcFace[34] embeddings. For long videos this thesis uses a slow-fast adaptation: a slow component fused within 2-second clips and a fast component comparing first-frame embeddings across clips, so that cross-clip identity drift is measured rather than averaged away.

A scope caveat applies to everything VBench-derived in this thesis: **neither VBench nor VBench-2.0 officially supports long-video super-resolution evaluation.** VBench-2.0 targets prompt-conditioned text-to-video generation (thirteen of its eighteen dimensions require text prompts and are inapplicable to SR), and the long-video mode used here is the `vbench2_beta_long` extension — a beta add-on to the VBench-1.x dimension suite, not a released VBench-2.0 configuration. All long-video VBench results in this thesis therefore rest on documented adjustments to the published method: the slow-fast Human_Identity adaptation (2-second clips, within-clip and cross-clip-first-frame branches), custom-input operation without prompts, and the beta long protocol for the consistency dimensions.

Beyond that scope adaptation, engineering care was required that is part of the evaluation's reproducibility story: six patches to the upstream VBench-2.0 stack (multi-face frames, late reference-frame initialisation, a zero-face division guard, and packaging/configuration fixes), and an fps-override correction — both evaluated SR pipelines hard-code their output frame-rate tag regardless of the source fps, which silently shifts clip boundaries in any per-second protocol and biased the cross-clip identity branch until corrected (the mechanism is characterised in Chapter 3).

2.4 The DOVE Evaluation Protocol

All ground-truth comparisons in this thesis follow the DOVE[13] protocol: low-resolution inputs are produced by the RealBasicVSR[18] degradation pipeline (bicubic downscale by a factor of four, sensor noise injection, codec compression), and full-reference metrics are computed in the DOVE RGB convention. The DOVE test suites used here are UDM10[7] and SPMCS[8] with realistic degradation, and YouHQ40[2] for the resolution-extrapolation ladder of Chapter 6. Validating against DOVE's published numbers anchors every experiment chapter: conclusions are drawn only on top of a numerically verified inference environment.

Chapter 3.　Failure Modes of Long-Video Super-Resolution Evaluation

This chapter documents empirically the four structural breakdowns of long-video SR evaluation summarised in Section 1.1: temporal-blind per-frame metrics, scale-blind short-range temporal metrics, consistency-erasing perceptual aggregates, and unsurfaced content-dependent calibration shifts. Each breakdown is established on a numerically validated evaluation environment, described first.

3.1　The Real-SR Test Set and Baseline Validation

A test set of five long-form synthetic super-resolution videos was prepared, each at 1280×720 output resolution, with total frame counts ranging from 2,412 to 5,000 per video and durations from 80 to 208 seconds at native frame rates. The total frame count across the set is 22,412. Each test video was generated by applying the DOVE[13] degradation pipeline (bicubic downscale by a factor of four, sensor noise injection, and codec compression) to a real source clip, and then upscaled by two representative super-resolution baselines: a detail-preserving diffusion-based method (MGLD-VSR[1]) and a text-conditioned temporal-diffusion smoother (Upscale-A-Video[2]).

Before drawing conclusions from any metric values, the inference environment was validated against published DOVE benchmark numbers: the detail-preserving baseline reproduces the published UDM10 results exactly to four decimal places (PSNR, SSIM, LPIPS, DISTS); the smoother baseline reproduces them within a consistent positive offset ($+1.33$ dB on UDM10, $+1.68$ dB on SPMCS) attributable to a minor degradation-pipeline configuration difference rather than a setup defect. All findings in this thesis are therefore drawn on top of a numerically validated baseline pair.

A subsidiary characterisation is the discovery and correction of a frame-rate metadata inconsistency in both super-resolution pipelines: each baseline tagged its output at a hard-coded frame rate independent of the source frame rate, which introduced a small but systematic bias in any per-second temporal aggregation. The correction is incorporated into all subsequent measurements.

3.2 A Content-Dependent Regime Shift in VBench-2.0 Metrics

Two slow-fast-adapted VBench-2.0 metrics — Human_Anatomy (anomaly classifier on per-frame body / face / hand detector ensemble) and Human_Identity (RetinaFace + ArcFace face embedding stability across the video) — were evaluated on the real-SR test set. Both metrics were adapted to long videos through a 2-second sliding-clip protocol with slow (within-clip) and fast (cross-clip first-frames) components fused.

On four of the five videos, both metrics rank the detail-preserving diffusion baseline higher than the text-conditioned smoother baseline, in agreement with informal viewing. On the fifth video — a close-up scene where the smoother method's output was visually substantially worse — **both metrics flip rankings in disagreement with the visual evidence**. Two independent metrics failing in the same direction on the same video, on the one video where the visual judgement is most confident, cannot be dismissed as evaluation noise.

Per-frame Anatomy fire-rate traces decomposed by detector channel and anomaly threshold identify the fraction of frame area occupied by the largest detected hand bounding box as the regime-shift trigger. Across the five-video cohort the median hand bounding-box fraction is approximately 2–3% of frame area; on the regime-shift video it is approximately 18%. The anomaly classifier was trained on a video distribution where close-up hands are rare; on close-up content it fires more aggressively, and it fires more aggressively on sharper output because sharper edges produce stronger detector activations. The classifier therefore inverts on close-up content: the visually better method is flagged as more anomalous by the trained behaviour the classifier is supposed to provide. Human_Identity exhibits a parallel inversion on the same video, likely through face-detection-confidence cliffs at close range.

The generalised lesson is that VBench-style anomaly-detector metrics have content-dependent calibration shifts that the metric does not surface. A video whose content happens to push the classifier into the high-fire regime will see every method evaluated on it scored under different effective calibration than other videos. There is no per-video reliability signal in the standard pipeline that flags "this video is out of the metric's calibrated range." The proposed metric (Chapter 4) addresses this by introducing per-sub-metric reliability gates that downweight any sub-metric whose calibration assumptions are violated on the video at hand.

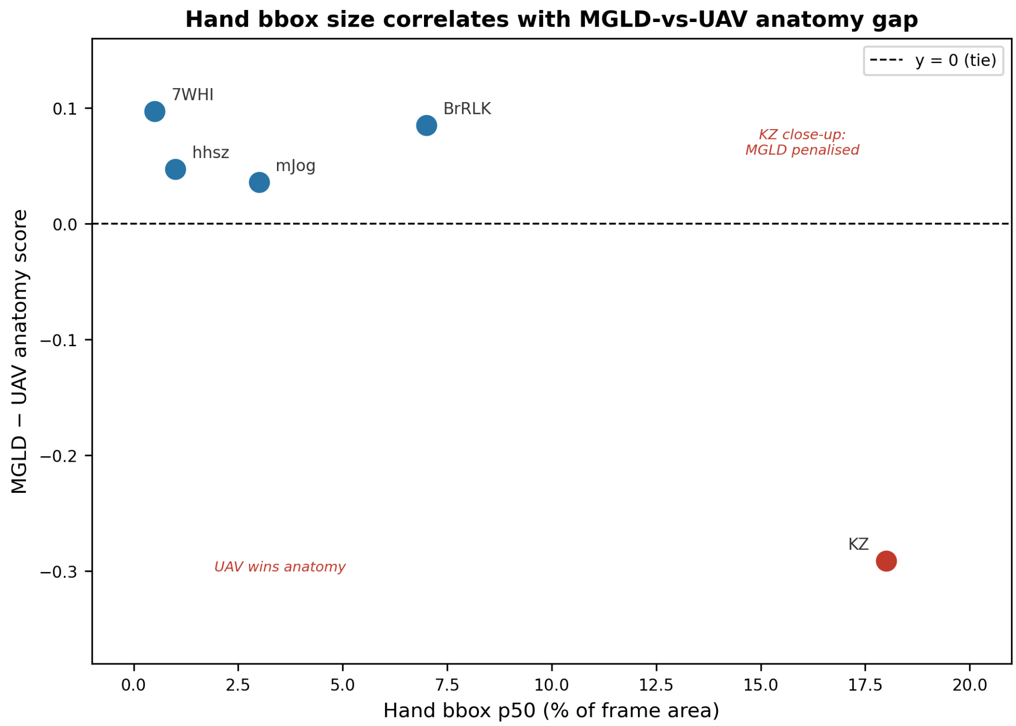

Figure 3.1 Anatomy metric difference between the detail-preserving and the text-conditioned smoother baselines, plotted against per-video median hand-bounding-box fraction. Four videos cluster in the upper-left at low close-up fraction, with the detail-preserving baseline correctly winning. The regime-shift video sits alone in the lower-right at approximately 18% hand-bbox fraction — well above the cohort median of 2–3% — with the metric inverted by approximately -0.29. The pattern is consistent with a content-dependent calibration shift triggered by close-up scale.

3.3 The tOF Temporal-Scale Crossover

The tOF metric[12] measures temporal consistency through optical-flow warping error and is conventionally evaluated at adjacent-frame lag $k = 1$. The framework was extended to compute tOF and tLP at multiple lags $k \in \{1, 5, 10, 30, 60, 120\}$ using RAFT-based[32] optical flow with forward-backward consistency masking applied uniformly across all k.

The winner ranking flips with k. At $k = 1$ the text-conditioned smoother baseline wins on four of five videos, because adjacent-frame transitions of a spatially smoother output have mechanically lower warping error. At $k \geq 30$ the detail-preserving diffusion baseline wins on four of five videos, because the smoother method's per-frame smoothness comes at the cost of cumulative drift across longer time horizons — drift that long-range warping picks up directly. The crossover point lies at $k \approx 5$–10; tLP exhibits the same crossover at the same range.

The interpretation is that short-range temporal metrics suffer from the smoother-output bias in its purest form: a smoother output has less to differ from the previous frame, so frame-to-frame difference is mechanically lower. This same bias affects DOVER's aesthetic branch, DINOv2-cosine cross-frame similarity, and the Anatomy fire-rate. The literature convention of reporting tOF at $k = 1$ as the temporal-consistency proxy is, in effect, a spatial-smoothness measurement.

The practical consequence is that a study that claims a temporal-consistency improvement using tOF at $k = 1$ may be measuring spatial smoothness, not consistency. A method that wins on the standard benchmark by lowering tOF at $k = 1$ may lose to its competitor at $k = 60$. Neither metric is correct in isolation; the picture only resolves when k is varied. The proposed metric (Chapter 4) aggregates tOF across $k \in \{1, 5, 10, 30, 60, 120\}$ with a weighting that gives meaningful representation to every temporal scale.

3.4 A Synthetic Artefact Battery Exposes Categorical Blind Spots

To establish the empirical reach of the limitations above, a parameterised synthetic artefact pipeline was constructed that operates on existing super-resolved outputs and injects controlled degradations at known severities. Four artefact families were implemented in the initial battery (the full twelve-family benchmark used for metric validation is presented in Chapter 5):

- **Colour drift** — a linear ramp: at frame i of T total frames, the red channel is multiplied by $1 + \alpha \cdot i/T$ and the green and blue channels are reduced by the same factor. Severity $\alpha \in \{0.02, 0.05, 0.10, 0.20, 0.40\}$. Tests detection of slow monotonic drift across the video.

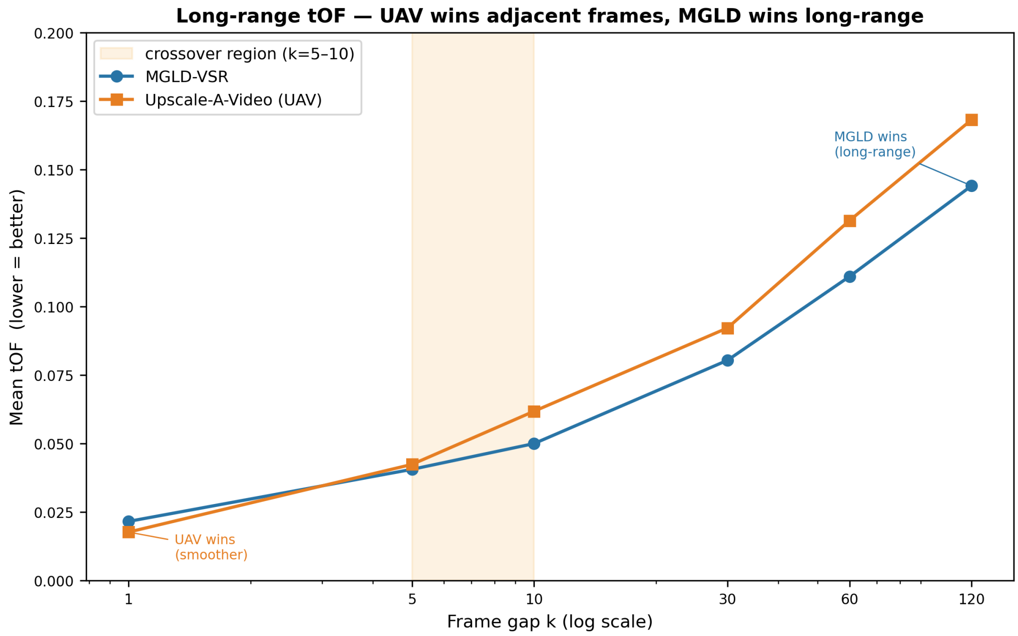

Figure 3.2 Mean tOF as a function of frame gap k for the two super-resolution baselines. At $k = 1$ the text-conditioned smoother baseline (orange) scores lower, consistent with the smoother-output bias of adjacent-frame temporal metrics. At $k \geq 30$ the detail-preserving baseline (blue) scores lower, registering the smoother baseline's cumulative long-range drift. The crossover region (shaded) lies at $k = 5$–10.

- **Chunk-boundary jumps** — at every 60th frame ($\approx$ two seconds at 30 fps), a deterministic per-chunk uniform additive offset of magnitude $\pm\alpha \cdot 255$ is applied to all channels. Severity controls step amplitude. Mimics the seams produced by tile-based diffusion super-resolution.

- **Periodic flicker** — sinusoidal brightness modulation: per-frame intensity is multiplied by $1 + \alpha \sin(2\pi i/P)$ with period $P = 15$ frames. Severity controls peak amplitude. Mimics the periodic flicker produced by some diffusion samplers.

- **Identity degradation** — per-frame frontal-face detection followed by a Gaussian blur with $\sigma = 10\alpha$ applied within each detected face bounding box. Severity controls blur sigma. Mimics the identity-collapse failure mode of diffusion super-resolution on faces.

Each generator is verified by unit tests checking that severity zero leaves the video unchanged within floating-point tolerance, that per-frame statistics match the analytic target curve, and that the construction is seed-reproducible.

Eight baseline metrics from the super-resolution literature were evaluated on the artefact videos generated from the two base videos available at this stage: CLIP-IQA, tOF at $k = 1$, tOF at $k = 60$, tOF at $k = 120$, tLP at $k = 120$, E*warp, DOVER, and Identity (fused slow-fast). For each (metric, artefact family) cell, the test is whether the metric responds monotonically across the five severities on both base videos.

Of the resulting 32 cells (8 metrics $\times$ 4 artefact families), only **five cells are clean PASS**. The verdict matrix is presented in Table 3.1.

Table 3.1 Severity-response verdicts of eight widely-used baseline metrics across four synthetic artefact families. PASS denotes monotonic severity-response on both base videos; PARTIAL denotes monotonic on one base or response only as a global side-effect; FAIL denotes flat or non-monotonic response.

Metric	Colour drift	Chunk-boundary	Flicker	Identity deg.
CLIP-IQA	FAIL	FAIL	FAIL	PARTIAL
tOF ($k = 1$)	FAIL	FAIL	**PASS**	FAIL
tOF ($k = 60$)	FAIL	**PASS**	FAIL	FAIL
tOF ($k = 120$)	FAIL	**PASS**	FAIL	FAIL
tLP ($k = 120$)	FAIL	**PASS**	FAIL	FAIL
E*warp	FAIL	**PASS**	FAIL	FAIL
DOVER	FAIL	FAIL	FAIL	FAIL
Identity (fused)	FAIL	FAIL	FAIL	PARTIAL

Four headline findings follow from this matrix.

Finding 1 — Colour drift is a categorical blind spot of the entire baseline metric suite. No baseline metric responds monotonically across the severity range on either base video. The mechanism is structural: a slow linear colour ramp produces a per-pair colour shift between frames k apart of $\alpha k/T$. For the conditions tested this per-pair shift sits below the histogram bin width of conventional colour-histogram metrics, below the noise floor of optical-flow warping error (the flow estimator absorbs uniform colour shifts), and below the perceptual threshold of CLIP-IQA and DOVER (the per-frame quality of a slightly red-shifted frame is indistinguishable from the original). The signal exists in the trajectory of per-channel means across the entire video, not in any per-pair comparison.

Finding 2 — Chunk-boundary detection requires temporal lag k at least as large as the chunk size. All four long-range temporal metrics catch chunk-boundary cleanly. Per-frame metrics, the adjacent-frame temporal metric, and the clip-level perceptual metrics are entirely blind. This proves that a single-scale temporal metric, however well-tuned, will miss artefacts

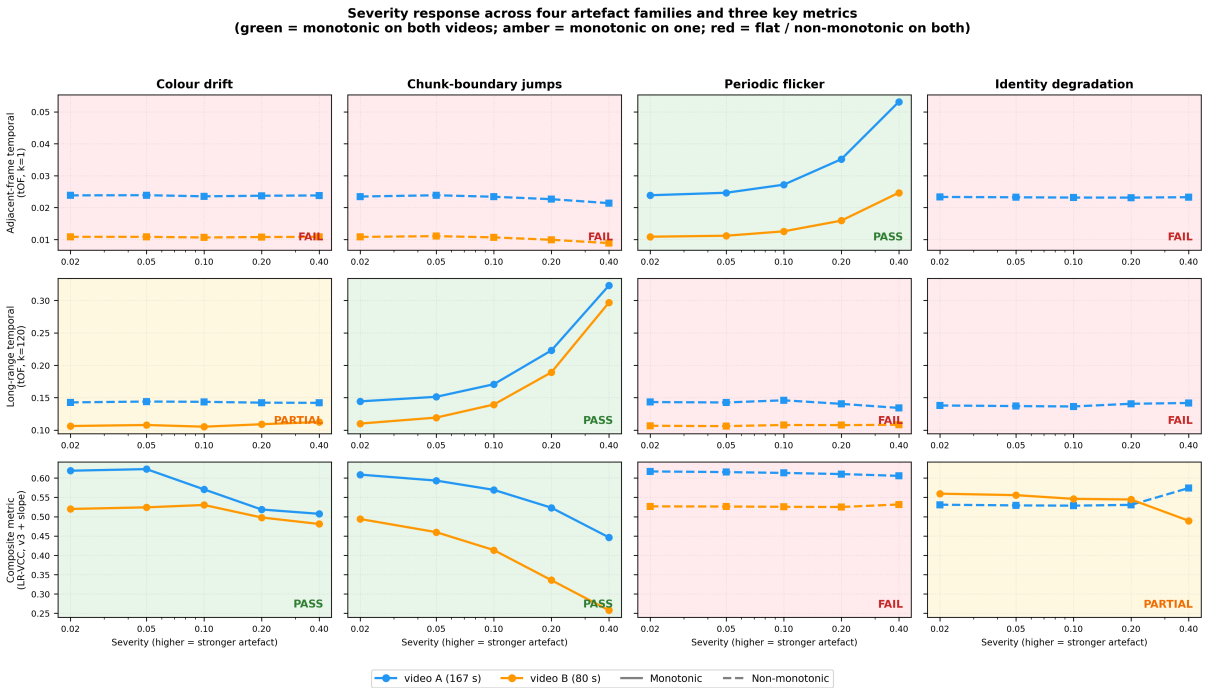

Figure 3.3 Severity response across the four artefact families and three representative metrics from the baseline suite: adjacent-frame temporal (tOF at $k = 1$), long-range temporal (tOF at $k = 120$), and an early five-sub-metric version of the proposed composite (Chapter 4). Each panel plots metric value against artefact severity on the two base videos. Green panels denote monotonic severity-response on both videos; amber panels denote monotonic on one video; red panels denote flat or non-monotonic response on both. No single baseline metric catches every artefact family, while the composite catches the artefact families whose mechanisms are addressed by its constituent sub-metrics and degrades gracefully on the others.

whose characteristic frequency does not match its measurement scale.

Finding 3 — Flicker is caught only at the smallest temporal lag. Long-range temporal metrics ($k \geq 30$) are blind to periodic flicker because the flicker period of 15 frames divides their measurement scale cleanly ($60 = 4P$, $120 = 8P$), so the per-pair difference cancels by construction. tOF at $k = 1$ catches it cleanly — the response is monotonic with severity ratio of approximately $4.5\times$ from the lowest to the highest severity. This is the converse of Finding 2: aggregation at long temporal scales alone is insufficient.

Finding 4 — Identity degradation triggers a content-dependent inversion of the slow-fast Identity metric. On the multi-face base video the Identity sub-metric responds correctly (score drop of approximately 0.23 across the severity range). On the single-face base video the same sub-metric rises with severity by approximately 0.11, because heavy face-region blur ($\sigma = 4.0$ at maximum severity) erases identity-distinctive features in all frames of the single face, so cross-clip first-frame embeddings become more similar in a "generic blurred face" sense. The face detector survives the blur (face detection rate of approximately 0.96 at every severity), so face-rate-based reliability gating does not engage. The pathology lies in the slow-fast pooling itself.

3.5 Why the Gaps Are Structural

The blind spots above are not artefacts of a particular threshold choice or training-data deficit. Each has a structural cause: per-frame metrics have no temporal awareness by construction; adjacent-frame temporal metrics exhibit the smoother-output bias by construction; clip-level perceptual aggregates remove the cross-clip signal by construction; per-pair temporal at a fixed k is scale-blind to artefacts whose characteristic frequency does not match k; and VBench-style anomaly-detector metrics have content-dependent calibration shifts intrinsic to learned classifiers operating on a wider content distribution than they were trained on.

Progress requires (a) multi-scale temporal sampling that gives weight to both high-frequency and long-range failure modes, (b) an explicit colour-trajectory sub-metric that catches drift slower than any per-pair measurement scale, (c) reliability gating per sub-metric so that content-dependent metric flips suppress themselves on the videos where they apply, and (d) a composition layer that downweights unreliable sub-metrics on-the-fly rather than reporting a single fixed metric. These four requirements specify the design space for any proposed long-video consistency metric; the metric introduced in Chapter 4 satisfies each.

Chapter 4. The Long-Range Video Consistency Composite

This chapter specifies the proposed Long-Range Video Consistency Composite (LR-VCC) metric, its design rationale, and the iterative protocol under which the production version was derived. The metric is constructed directly from the failure modes characterised in Chapter 3 — plus one further failure mode discovered during validation itself (Section 4.2); each sub-metric closes a specific gap, and the composition layer downweights any sub-metric whose calibration assumptions are violated on the video under evaluation.

4.1 Architecture

Seven sub-metrics, indexed by $s \in \{A, T, I, D, D', D'', E\}$, are computed in parallel from a super-resolved video V:

- **Sub-metric A — Appearance stability.** Per-frame CLIP-IQA[23] is computed over the video. The score is $\mathrm{mean}(q) - \lambda \cdot \mathrm{std}(q)$ with $\lambda = 0.5$, rewarding outputs that are not only high-quality on average but consistently so across minutes of footage. Reliability is gated on two regimes in which the signal stops discriminating: a near-constant quality trace ($\mathrm{std}(q) < 0.02$) and quality saturation ($\mathrm{mean}(q) > 0.98$).

- **Sub-metric T — Temporal stability.** Optical-flow warping error (tOF[12]) is computed at multiple temporal lags $k \in \{1, 5, 10, 30, 60, 120\}$ using RAFT-based optical flow with forward-backward consistency masking. The aggregate is a weighted mean over k; uniform weighting is adopted as the production setting because the empirical temporal-scale crossover (Section 3.3) shows that no single k is sufficient. The score is one minus the (normalised) weighted mean warping error, where lags whose RAFT forward–backward mask coverage falls below 0.10 are excluded; reliability is the mean, over all lags, of the smooth coverage gate — a video whose long-lag flow is mostly unreliable yields a low-confidence temporal signal rather than a spurious score.

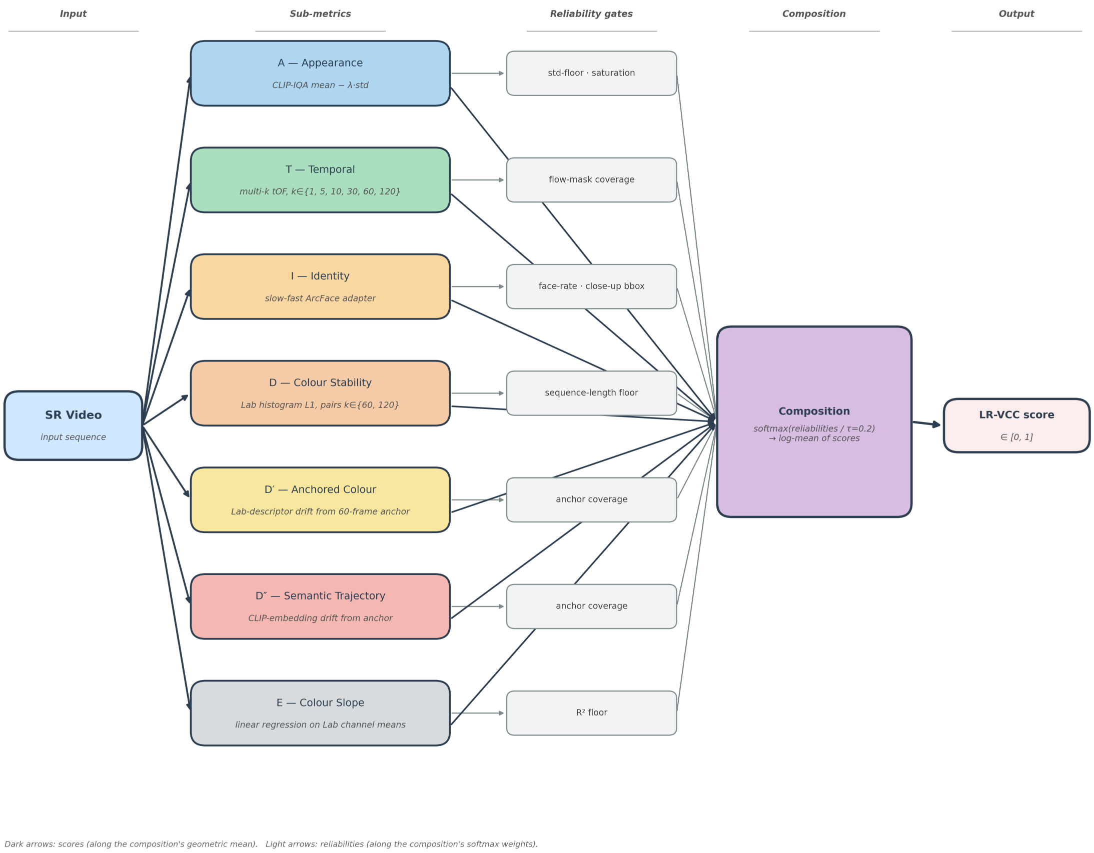

Figure 4.1 LR-VCC v5 architecture: a super-resolved video as input; seven sub-metrics computed in parallel, each emitting a (score, reliability) pair in $[0,1]\times[0,1]$; a reliability-weighted softmax-log-mean composition produces the final LR-VCC score in $[0,1]$. Dark arrows carry scores (the geometric-mean path); light arrows carry reliabilities (the softmax-weight path).

- **Sub-metric I — Identity preservation.** The slow-fast Human_Identity adapter (RetinaFace + ArcFace face embedding stability across 2-second clips with both within-clip and cross-clip first-frame components) provides the score. Reliability is a product of independent gates: a face-detection-rate gate (below a 0.20 clip rate there is no identity signal) and a close-up gate on the median face/hand bounding-box fraction (threshold 0.05 of frame area — the trigger identified in Section 3.2), which downweights the sub-metric precisely on the content where the underlying classifiers exhibit calibration shifts. A third, dispersion-based gate (abstaining when per-clip scores are erratic) is implemented but ships disabled: its threshold could not be calibrated defensibly on two single-face base videos (separation 0.003), so it is parked pending a larger calibration set.

- **Sub-metric D — Colour stability (pairwise).** Per-frame Lab-channel histograms are compared at L1 distance over multi-scale frame pairs $k \in \{60, 120\}$. The score is $\exp(-\alpha \cdot \overline{d}_{\text{hist}})$ with α calibrated on chunk-boundary empirics. Reliability is gated by sequence length (at least 240 frames are required for the $k{=}120$ pairs to be meaningful). D is retained in v5 alongside its anchored replacements because pairwise stability still carries complementary signal on boundary-type artefacts.

- **Sub-metric D′ — Anchored colour stability.** Each frame is summarised by a Lab-space descriptor (a 32-bin per-channel histogram concatenated with normalised per-channel means, giving sub-bin sensitivity to small shifts). The anchor is the mean descriptor over the first 60 frames ($\approx$ 2 seconds); every subsequent frame contributes its L1 distance to the anchor. The distance sequence is summarised by quarter means $q_1, \ldots, q_4$, and the score is an exponential decay in the trajectory's net drift, $\exp(-\beta_{D'}\,|q_4 - q_1|)$. Measuring drift away from a fixed anchor rather than between frame pairs closes the convergence-rewarding pathology of sub-metric D (Section 4.2): a corruption that makes frames converge toward a static appearance still walks away from the opening anchor.

- **Sub-metric D″ — Semantic trajectory stability.** The same anchored-trajectory construction as D′, but in CLIP[31] image-embedding space (ViT-B/32, one embedding per 8th frame) with cosine distance in place of histogram L1: the anchor is the mean embedding of the first 60 frames, and the score is $\exp(-\beta_{D''}\,|q_4 - q_1|)$ over the quarter means of the anchor-distance sequence. Because CLIP embeddings respond to scene content rather than colour statistics, D″ registers semantic drift invisible to histogram measures — progressive background replacement with a stable colour distribution, or the slow content drift of a streaming super-resolver — and is robust to the camera motion that perturbs D′.

- **Sub-metric E — Colour trajectory slope.** Per-frame Lab-channel means are fit by linear regression as a function of frame index. The score is an exponential decay in the maximum absolute slope across channels. Reliability is the maximum coefficient of determination (R^2) across the three channels, gated by an R^2 floor. A drifting video has high R^2 (the slope is real, so the metric speaks confidently); a flickering video has R^2 near zero (sinusoidal residuals dominate, so the metric correctly abstains); a clean video has near-zero slope (the score is near one regardless of reliability).

Each sub-metric outputs a score $s_i \in [0, 1]$ and a reliability $r_i \in [0, 1]$. The composition is a reliability-weighted log-mean:

$$w_i = \frac{\exp(r_i/\tau)}{\sum_j \exp(r_j/\tau)}, \tag{4-1}$$

$$\text{LR-VCC}(V) = \exp\left(\sum_i w_i \log(s_i + \varepsilon)\right), \tag{4-2}$$

with temperature $\tau = 0.2$ (a sharp softmax: the most reliable sub-metric dominates) and $\varepsilon = 10^{-6}$ for numerical stability. A video for which all reliabilities fall below a low-confidence threshold is flagged in the output and excluded from per-method aggregates by default.

The log-mean composition preserves the no-compensation property across sub-metrics: a sub-metric scoring near zero pulls the composite down unless the other sub-metrics carry overwhelming weight, which in turn requires the failing sub-metric's reliability to be near zero. A weighted arithmetic mean would not have this property: a single sub-metric failure could be diluted away by averaging.

Table 4.1 lists the canonical configuration. All decay constants were calibrated on characterised regimes (chunk-boundary empirics for α; colour-drift severity response for β_E), not tuned to validation outcomes; the sensitivity of every headline result to these choices is quantified in Chapter 5.

4.2 Design Iterations: Closing Characterised Failure Modes

LR-VCC was developed under an explicit protocol: every design iteration must close a specific failure mode characterised on the validation set, not apply an unmotivated parameter tweak. The version history is:

v1 (A, T, I). Three sub-metrics with per-video reliability tests. Passed perceptual-ordering validation on the real-SR test set but blind to colour-drift artefacts.

Table 4.1 Canonical LR-VCC v5 configuration.

Parameter	Value	Role
λ	0.5	A: std-deviation weight
temporal weighting	uniform over k	T: no single scale suffices
α	0.394	D: histogram-distance decay
β_E	200	E: slope decay
$\beta_{D'}$	0.5	D′: anchor-drift decay
$\beta_{D''}$	3.0	D″: semantic-drift decay
τ	0.2	composition softmax temperature
ε	10^{-6}	log-mean stabiliser
low-confidence floor	0.2	all-reliabilities-below flag

v2 (+D). Added the pairwise colour-histogram sub-metric to target colour drift; detection remained incomplete.

v3. Recalibrated D's exponential decay constant against chunk-boundary empirics.

v3+slope (+E). Added the trajectory-slope sub-metric with R^2-gated reliability, closing the colour-drift blind spot (Finding 1 of Section 3.4).

v4. Scaled validation from two to five base videos and from four to six artefact families (adding progressive identity drift and progressive background replacement). The expanded verdict matrix revealed a new pathology: **sub-metric D systematically rewards convergence-type corruptions**. Replacing a dynamic background with a static reference makes per-frame colour distributions more stable over time, so a pairwise-stability score improves as the corruption gets worse; the softmax composition then amplifies D's confident, perverse signal. Background-drift rankings inverted on four of five base videos.

v5 (+D′, +D″; production). Replaced pairwise stability with anchor-referenced measures on two complementary channels: colour statistics (D′) and CLIP semantics (D″). Background-drift inversions dropped from four of five bases to one (a cartoon-content base, with halved magnitude), and a six-transform flip ablation family was added as a designed control battery for sub-metric attribution (Chapter 5).

The v4 lesson generalises beyond this metric: stability-oriented consistency metrics have a structural failure class on convergence-type corruptions, where the corrupted video is more "stable" than the clean one. Anchor-referenced trajectory measures are the systematic fix.

4.3 Reliability Gating as the Core Design Pattern

The central design hypothesis is that every existing learned-representation video metric is reliable on some content and unreliable on other content; rather than discard failed metrics or attempt to train new bias-free metrics, the composition uses existing metrics honestly and controls their influence per video through reliability scores derived from the regime indicators that predict their failure mechanisms.

Each per-sub-metric reliability is computed from regime indicators using smooth sigmoid penalties: an indicator value v against a threshold t incurs penalty $\sigma\big(s \cdot (v - t)\big)$ (or its mirror for below-threshold regimes), with sharpness $s = 10$ producing smooth transitions rather than hard cliffs, and reliabilities formed as products of $(1 - \text{penalty})$ factors. The thresholds were selected from the regime characterisations in Chapter 3; they are not tuned to the validation set.

A sharp softmax over reliabilities ensures that when one sub-metric is substantially more trustworthy than the others on a given video, that sub-metric clearly dominates the composition. This is the mechanism that makes the composite resistant to the content-dependent ranking flips of Section 3.2: on a video in the close-up regime, where the Identity-pathway anomaly classifier is known to misbehave, the Identity sub-metric's reliability is downweighted by the close-up gate, allowing the other sub-metrics to determine the composite outcome.

4.4 Validation Strategy

The composite is validated at three layers; results are presented in Chapter 5.

Layer 1 — perceptual agreement on the real-SR test set. The composite must rank the detail-preserving baseline above the text-conditioned smoother on the per-method mean and on every per-video comparison.

Layer 2 — flip-resistance on the regime-shift case. The composite must not flip on the close-up video where Human_Anatomy and Human_Identity both flip against visual judgement.

Layer 3 — severity-response on the synthetic artefact benchmark. For each artefact family, the composite should respond monotonically to severity, with any non-monotonic cells attributable to a named failure mechanism rather than unexplained noise.

4.5 Reproducibility and Decomposability

The metric is fully decomposable. Each sub-metric is computed independently, and the result is cached per video. Hyperparameter studies and re-tuning of the per-sub-metric decay constants and weighting choices do not require re-scanning videos; the cached per-video values are re-aggregated through the composition layer in seconds. Additional sub-metrics can be added to the composition without disturbing the cached values of the existing sub-metrics.

The output of the composite on a given video is the LR-VCC score, the per-sub-metric $(\text{score}, \text{reliability})$ pairs, the softmax-derived sub-metric weights, the low-confidence flag, and a set of per-sub-metric diagnostic indicators (close-up indicator, face-detection rate, mean mask coverage at long temporal lags, regression R^2, histogram entropy). The per-video per-sub-metric matrix is always reported alongside the headline composite, so a reviewer can see where the composite is driven from on each video. There is no black-box aggregation.

Chapter 5. Experimental Validation

5.1 Experimental Setup

Base videos. Five long-form base videos (identified as 7WHI, BrRLK, KZ, hhsz, and mJog throughout), 80–208 seconds each, 2,412–5,000 frames, 22,412 frames in total, degraded by the DOVE pipeline (Section 2.4) to 320×180 and super-resolved to 1280×720.

Synthetic benchmark. Twelve artefact families at five severities over the five base videos (300 clips): the four families of Section 3.4 (colour drift, chunk-boundary, flicker, identity degradation), two progressive long-range families (identity drift, background drift), and a six-transform flip control family (horizontal, transpose, periodic, elastic, channel-shuffle, invert) designed for sub-metric attribution with predicted-blind and predicted-caught cells.

Evaluated methods. MGLD-VSR[1], Upscale-A-Video[2], FlashVSR[3] (Section 2.1), and — as a designed contrast — a frame-wise anchor: RealESRGAN[18] applied independently per frame, i.e. super-resolution with no temporal modelling at all, which any temporal-consistency metric should be able to tell apart from genuine video methods.

5.2 Synthetic Artefact Validation

For each (artefact family, base video) condition, the composite's severity-response is summarised by the score change from the lowest to the highest severity, $\Delta = \text{LR-VCC}(\text{sev } 0.40) - \text{LR-VCC}(\text{sev } 0.02)$, classified as pass ($\Delta \leq -0.05$), weak ($\Delta \leq -0.02$), flat ($|\Delta| < 0.02$), or inverted ($\Delta \geq +0.02$). Table 5.1 presents the full matrix, recomposed uniformly under the canonical configuration.

Four observations structure the reading of Table 5.1.

The colour-drift blind spot is closed. Colour drift — invisible to every one of the eight baseline metrics in Section 3.4 — is clean on all five bases (four pass, one weak), the direct payoff of sub-metrics E and D′.

Table 5.1 LR-VCC v5 severity-response verdict matrix: twelve artefact families × five base videos. Each cell shows the verdict and Δ (severity 0.02 → 0.40). Clean (pass+weak): 29/60 conditions. Source: `verdict_matrix_v5_uniform.md`.

Artefact	7WHI	BrRLK	KZ	hhsz	mJog
colour_drift	PASS (−.11)	PASS (−.06)	PASS (−.15)	WEAK (−.05)	PASS (−.11)
chunk_boundary	PASS (−.07)	INV (+.11)	PASS (−.10)	PASS (−.11)	WEAK (−.04)
background_drift	FLAT (+.00)	INV (+.05)	PASS (−.07)	PASS (−.20)	FLAT (+.00)
flicker	FLAT (−.00)	INV (+.06)	WEAK (−.03)	FLAT (+.01)	FLAT (+.00)
identity_degradation	INV (+.03)	FLAT (+.00)	WEAK (−.03)	PASS (−.05)	FLAT (+.01)
identity_drift	FLAT (−.00)	FLAT (−.00)	WEAK (−.03)	PASS (−.07)	FLAT (−.00)
flip_invert	PASS (−.15)	PASS (−.22)	PASS (−.35)	PASS (−.39)	PASS (−.07)
flip_channel_shuffle	WEAK (−.03)	PASS (−.08)	PASS (−.31)	FLAT (+.00)	PASS (−.06)
flip_transpose	FLAT (+.01)	FLAT (+.00)	PASS (−.05)	PASS (−.06)	WEAK (−.03)
flip_periodic	FLAT (−.01)	FLAT (−.02)	FLAT (+.00)	FLAT (−.02)	FLAT (−.01)
flip_elastic	FLAT (−.00)	FLAT (+.00)	FLAT (−.00)	FLAT (+.01)	WEAK (−.02)
flip_horizontal	FLAT (+.02)	FLAT (+.01)	FLAT (−.00)	FLAT (−.01)	FLAT (−.02)

The v5 anchored redesign fixed the convergence pathology. Under v4, progressive background replacement inverted on four of five bases — the composite rewarded the corruption. Under v5 a single inversion remains (BrRLK, with halved magnitude), with the two strongest responses (pass at -0.20 and -0.07) on the bases where the anchor is most stable. The two flat cells share a mechanism: on content with substantial natural scene change, the anchor distance saturates early and the injected drift adds little.

The flip control battery behaved as designed. The flip family was constructed with predicted outcomes: flip_invert (histogram-destroying positive control) should be caught everywhere — it is, pass on all five bases; flip_horizontal (histogram-preserving, CLIP-robust) should be invisible — it is, flat everywhere, a designed null result that confirms the sub-metrics measure what they claim rather than any change whatsoever; flip_channel_shuffle (marginal-preserving but appearance-breaking) should be caught partially — four of five clean. Periodic and elastic transforms, predicted mostly invisible, are mostly flat. The battery turns the verdict matrix from a scorecard into a falsifiable instrument check.

The residual failures are localised and mechanistic. The BrRLK column carries three of the five inversions: cartoon content with frequent hard scene cuts, where natural anchor-distance variation dominates any injected signal — a documented content-domain limitation (Section 5.6). The identity rows are the weakest families (the slow-fast pooling pathology of Section 3.4, Finding 4, plus the parked dispersion gate); flicker remains largely composite-invisible because the sub-metrics that respond to it (T at $k{=}1$) rarely dominate the softmax — both are characterised, named failure modes rather than unexplained noise.

5.3 Real-Model Evaluation

Applying LR-VCC v5 to the three methods on the five-video real-SR test set yields the ranking in Table 5.2. The canonical protocol applies the metric as designed: fps-corrected identity inputs, the close-up reliability gate active for every method, and the dispersion gate off (parked pending calibration data).

Table 5.2 LR-VCC v5 composite per video and method under the canonical (uniformly gated) protocol. Bold marks the per-video winner. RealESRGAN is the frame-wise (no temporal modelling) anchor. Source: `realmodel_v5_gated.md`.

Video	MGLD-VSR	UAV	FlashVSR	RealESRGAN
7WHI	**0.738**	0.700	0.737	0.736
BrRLK	**0.402**	0.379	0.393	0.401
KZ	**0.750**	0.705	0.722	0.724
hhsz	**0.566**	0.545	0.550	0.529
mJog	**0.654**	0.617	0.649	0.631
Mean	**0.622**	0.589	0.610	0.604

MGLD-VSR leads on every video and on the mean, though the margins against FlashVSR are small (7WHI and mJog are near-ties at 0.001 and 0.005); UAV is third throughout. The comparison is protocol-honest: an interim table composed without the close-up gate on two of the three methods ranked FlashVSR first (0.610 vs 0.589) — the FlashVSR-vs-MGLD comparison depends on how much weight the identity sub-metric receives, which is itself quantified in the ablations below. The MGLD $>$ UAV ordering, by contrast, is unconditional: it holds on every video under the canonical protocol and in 50 of 52 sensitivity configurations (Section 5.5).

Table 5.3 decomposes the composite. FlashVSR's closeness to the top is driven by identity preservation (0.598 vs MGLD's 0.557 and UAV's 0.459) — and FlashVSR is simultaneously worst on D″, the sub-metric most sensitive to long-range semantic drift from the anchor (0.862

vs 0.893/0.884). A streaming model whose temporal positions grow without bound, carrying one mid-video seam on four of five videos, is exactly the design one would suspect of long-range drift; whether position encoding is actually the cause is the question Chapter 6 answers. UAV's single win is the temporal cell (0.942) — the smoother-output bias of warping-error measures, visible even inside the composite, and the reason T's reliability rarely dominates the softmax.

Table 5.3 Per-sub-metric means over the five videos (canonical protocol). Bold marks the best method per sub-metric; underline marks the worst. Source: `realmodel_v5_gated.md`.

Sub-metric	MGLD-VSR	UAV	FlashVSR	RealESRGAN
A (CLIP-IQA appearance)	**0.447**	0.336	0.400	0.432
T (tOF temporal)	0.934	**0.942**	0.939	0.930
I (slow-fast identity)	0.557	0.459	**0.598**	0.448
D (histogram stability)	0.505	0.506	0.501	**0.508**
E (colour slope)	0.339	**0.341**	0.336	0.328
D' (anchor histogram)	**0.708**	0.687	0.705	0.699
D'' (CLIP trajectory)	**0.893**	0.884	0.862	0.867

The frame-wise anchor lands where it should. RealESRGAN's signature is the mirror image of a genuine video method: second-best per-frame appearance (its outputs are sharp) but the worst identity, temporal, and colour-slope cells of the four methods — per-frame processing flickers exactly where temporal modelling stabilises. Its composite therefore sits mid-pack: above UAV, whose per-frame quality deficit outweighs its smoothness advantage, and below both genuine temporal methods' best-performing configurations. The anchor also sharpens an honest boundary of the study: the composite separates the anchor from MGLD-VSR clearly on the temporal-family cells, but multi-scale tOF alone barely does (0.930 vs 0.934) — reinforcing that anchored drift and identity, not warping error, carry the discriminative temporal signal on well-behaved methods.

Relation to per-frame quality metrics. Per-frame no-reference quality measures agree with LR-VCC's MGLD-vs-UAV ordering — CLIP-IQA (0.496 vs 0.391), MUSIQ (65.1 vs 56.3), NIQE, BRISQUE, and DOVER (73.8 vs 65.1) all favour the detail-preserving method — but they carry no consistency signal at all: Chapter 3's battery shows them blind to every long-range artefact family, and they would score a colour-drifting or identity-collapsing video identically to a stable one. The division of labour is therefore sharp: per-frame metrics measure quality without consistency; the SOTA consistency dimensions measure a smoothness-confounded consistency that inverts against perception (Table 5.4); LR-VCC is the instrument that supplies a consistency signal which is simultaneously perception-aligned.

5.4 Comparison with the Closest Existing Benchmark

The nearest published instruments to LR-VCC's goal — the two state-of-the-art baselines for video consistency assessment — are VBench's[25] subject-consistency dimension (DINO-feature cosine similarity between adjacent frames and against the first frame) and background-consistency dimension (the same construction on CLIP features), aggregated per video via the long-video clip protocol. A protocol note for fairness: long-video SR input is not an officially supported configuration for these dimensions (Section 2.3) — they are run here through the `vbench2_beta_long` extension in custom-input mode, i.e. the published method with documented adjustments, at its default long-video weighting ($w_{\mathrm{inclip}}=1$, $w_{\mathrm{clip2clip}}=0$). Table 5.4 evaluates both on the real-SR test set alongside the degraded low-quality input itself.

Table 5.4 VBench consistency dimensions on the real-SR test set (five-video means), including the degraded LQ input as a row. Higher is better according to the benchmark.

Dimension	LQ input	MGLD-VSR	UAV
subject_consistency	0.8936	0.8927	**0.9031**
background_consistency	**0.9333**	0.9235	0.9317

Two readings disqualify these dimensions as long-video SR consistency measures, and both follow from the smoother-output bias characterised in Chapter 3. First, **background consistency ranks the degraded input above both of its super-resolutions** (and subject consistency places it above MGLD): a blurry, detail-poor video produces more stable DINO features than any faithful restoration of it, so the benchmark scores the degradation as an improvement.

A consistency measure that prefers the input to its restoration cannot evaluate restoration. Second, **both dimensions order UAV above MGLD-VSR** — the inverse of human judgement and of LR-VCC's ranking — because feature-space stability mechanically rewards the smoother method. LR-VCC's design premises (multi-scale measurement against real motion, anchored drift rather than pairwise stability, reliability gating) are precisely the deltas that separate it from this closest neighbour.

Severity response of the SOTA dimensions. The ranking inversion above could in principle be excused if the dimensions at least detected long-range corruption. They do not. Both dimensions were run over the background-drift and colour-drift artefact families — fifty full-length clips regenerated bit-reproducibly — under the identical $\Delta(\text{sev } 0.02 \rightarrow 0.40)$ verdict protocol as Table 5.1. The result is universal insensitivity (Table 5.5): **0 of 20 (dimension, family, base) cells respond** — every cell is flat, including background-consistency on background drift, the dimension's namesake failure mode — and 14 of the 20 sub-threshold drifts run in the rewarding direction. LR-VCC is clean on 7 of the same 10 (family, base) conditions.

Table 5.5 Severity response of the SOTA consistency dimensions on the long-range artefact families (five bases each), versus LR-VCC on the same clips. Full per-base tables: `vbench_sota_verdicts.md`.

Dimension	Artefact	Clean	Δ range	LR-VCC clean
background_consistency	background_drift	0/5	$[-0.012, +0.004]$	2/5
background_consistency	colour_drift	0/5	$[+0.000, +0.006]$	5/5
subject_consistency	background_drift	0/5	$[-0.010, +0.010]$	2/5
subject_consistency	colour_drift	0/5	$[+0.000, +0.005]$	5/5

The mechanism is designed in, not incidental. The long-video protocol these dimensions ship with weights the cross-clip component to zero ($w_{\text{clip2clip}}=0$): the reported score is within-2-second-clip feature consistency, re-aggregated. A corruption that develops across clips — colour ramping over minutes, a background walking toward a replacement — changes each 2-second window almost nothing and is therefore invisible by construction. This is the empirical form of the structural argument in Chapter 3: clip-level re-aggregation removes exactly the signal that long-video consistency assessment requires. Combined with the ranking result above — degraded input scored above its restorations — the two SOTA dimensions fail both requirements of a long-video consistency measure on this benchmark: they do not rank perceptually, and they do not detect.

5.5 Ablations and Sensitivity

Because every sub-metric caches its per-video measurements, the composite can be recomposed under alternative hyperparameters or with sub-metrics removed in seconds, without re-scanning any video. Two studies exploit this.

5.5.1 Hyperparameter Sensitivity

A 52-configuration grid varies the anchored-drift decays $\beta_{D'} \in \{0.25, 0.5, 1, 2\}$ and $\beta_{D''} \in \{1, 2, 3, 5\}$ crossed with the softmax temperature $\tau \in \{0.1, 0.2, 0.5\}$, plus one-at-a-time variations of α and β_E. Across the grid: the per-video MGLD $>$ UAV ordering — the metric's core perceptual-agreement claim — holds on every video in 50 of 52 configurations; the three-method mean order is unchanged in 45 of 52; and the verdict matrix's clean count stays within $[26, 32]$ of the production 29/60, with the cells that change class concentrated in already-borderline weak/flat boundaries. All seven order changes occur at $\tau = 0.5$: softening the softmax dilutes the reliability weighting that the design depends on (Section 4.3), collapsing the composition toward an ungated mean. Sharp reliability weighting is thus a load-bearing choice, not a free parameter — and within any reasonable neighbourhood of the production temperature, the headline results are stable.

5.5.2 Leave-One-Out Sub-Metric Ablation

Dropping each of the seven sub-metrics in turn and recomposing tests hypothesis H1 directly: does the composite catch what no constituent catches, and does each sub-metric earn its place?

- **Each colour-family sub-metric fails in its designed direction when removed.** Dropping D′ flips background-drift cells to inverted (the convergence pathology returns); dropping E degrades the colour-drift row; dropping D″ or D moves 10–13 cells each, concentrated in the drift and boundary families they respectively target. No single sub-metric's removal leaves the matrix intact — the composition is load-bearing, not redundant (H1 supported).

- **Dropping identity flips the FlashVSR-vs-MGLD comparison.** FlashVSR's proximity to the top of the real-model table rests on the identity sub-metric; with I removed, MGLD leads by a wide margin. Combined with the closeup-gating analysis above, the honest summary is that MGLD-vs-FlashVSR is an identity-weighted comparison, while MGLD $>$ UAV holds under every ablation.

- **Dropping temporal changes nothing** — neither the real-model ordering nor a single verdict cell. On these surfaces T rarely dominates the softmax, consistent with its near-tied scores across well-behaved methods (Table 5.2); its value is diagnostic (the k-sweep analysis of Chapter 3) rather than discriminative within the composite.

5.5.3 Composition Provenance Audit

Recomposing from cached measurements also enabled a full reproduction audit of every published composite. The audit reproduced the production outputs bit-exactly and surfaced three composition-era inconsistencies in interim tables (a reliability-formula change mid-campaign, a close-up gate applied to only one method, and a stale identity input for one method), all corrected in the canonical protocol used throughout this chapter. The uniform recompose shifts the clean count from 28/60 to 29/60 and changes six cell classes — none affecting any headline claim. The episode reinforces a reproducibility practice this thesis adopts throughout: cache raw measurements, re-derive scores, and gate every published table on bit-exact reproduction.

5.6 Limitations

Every residual failure in this chapter has a named mechanism, recorded here both as honest scope statements and as future-work targets.

Cartoon / scene-cut-heavy content. The BrRLK base carries most of the matrix's inversions. Its frequent hard scene cuts mean the opening-window anchor of D′ and D″ does not represent the video: natural anchor-distance variation dominates injected corruption. An adaptive, scene-cut-aware anchor is the natural fix (Chapter 7).

Pure horizontal mirroring is invisible. flip_horizontal preserves colour marginals by construction and CLIP embeddings are documented to be robust to horizontal flips, so all seven sub-metrics are blind to it — a designed-in null result documented as a metric limit. A mirror-sensitive sub-metric (pixel-correlation or pose-based) would close it.

Identity families remain the weakest rows. The slow-fast pooling pathology (Section 3.4, Finding 4) is only partially mitigated: the dispersion gate designed for it could not be calibrated defensibly on two single-face bases (separation 0.003) and ships parked. Progressive identity drift in particular responds on only two of five bases.

Scale of the benchmark. Five base videos and three (soon five) SR methods support the mechanism-level claims made here, but community-benchmark claims would need an order of magnitude more content and human-judgement calibration — deliberately deferred (Chapter 7).

Chapter 6. RoPE Extrapolation in Streaming Diffusion Video Super-Resolution

6.1 Motivation and Research Questions

The real-model evaluation of Chapter 5 left a pointed question open: FlashVSR — within 0.012 of the top composite score and the outright winner on identity preservation — exhibits the worst long-range semantic drift (sub-metric D″) of the three methods — and FlashVSR is precisely the method whose streaming design lets its temporal RoPE index grow without bound as the video lengthens. Its stock frequency table caps a single pass at 4,089 frames ($\approx$ 2.3 minutes), and its trained temporal window covers only 81 frames (21 latent frames after $4\times$ VAE compression); its trained spatial extent is 48 latent rows (from the 768×1408 distillation resolution). Position-encoding extrapolation is therefore the natural suspect for long-video degradation, and the same suspicion extends to the spatial axes at high resolution.

The study decomposes positional extrapolation into two falsifiable hypotheses:

H0 (Magnitude). Absolute position offsets beyond the trained window degrade quality.

H1 (Geometry). Distortion of relative-distance geometry (dilation or compression of position spacing) degrades quality.

The decomposition matters because streaming inference stresses exactly the magnitude axis (indices grow forever) while window extension stresses the geometry axis (more positions inside one attention span). Which one costs quality determines where engineering effort should go.

6.2 The Position-Injection Instrument

A runtime position-injection hook swaps any one of the three RoPE frequency tables (t/h/w) — a duck-typed replacement of the DiT's per-axis frequency-table attribute — for a wrapper that transforms baseline positions: shift ($p \mapsto p + k$), stretch ($p \mapsto s \cdot p$), continuous-position interpolation, and modulo bounding, with on-demand table extension past the stock 1,024 rows. Continuous interpolation evaluates the RoPE formula directly at fractional positions rather than interpolating table rows (averaging complex phases would be incorrect); evaluated at integer positions it reproduces the stock table's effect bit-for-bit, providing an identity condition at the numerics floor.

The instrument satisfies three faithfulness gates: (i) bit-exact when idle — with the hook installed but no transform active, output drift against stock is 0.0 on a measured 0.0 nondeterminism floor; (ii) engagement-checked — every run asserts the transform actually fired; (iii) repository-pristine — zero modification of the tagged FlashVSR codebase, with 48 unit tests covering the hook. Content and compute are identical across conditions; only the position labels change, which is what makes the comparisons causal.

6.3 Experimental Design

Seven experimental arms cover the two hypotheses per axis: (1) temporal shift sweeps up to position 996; (2) temporal stretch sweeps, integer and continuous, $s \in [0.5, 250]$; (3) ground-truth scoring on DOVE-UDM10 (10 clips $\times$ 29 frames, realistic degradation; PSNR/SSIM/LPIPS via pyiqa in the DOVE RGB convention, on the protocol whose stock score — 24.02 dB beside MGLD's verified 24.23 — anchors every number); (4) the D$''$ causal check on the 2,412–5,000-frame real-SR videos: stock segmented vs true single-pass vs magnitude-bounded (positions cycled mod 336) arms; (5) spatial (h/w) label sweeps on UDM10; (6) a real resolution-extrapolation ladder on YouHQ40 (8 clips, 1080^2 GT): latent grids $48^2 \rightarrow 72^2 \rightarrow 80^2 \rightarrow 88^2 \rightarrow 96^2$, i.e. $1\times$ to $2\times$ the trained spatial extent (input padding quantises grids to multiples of 8), stock vs spatial-PI positions; and (7) collapse-decomposition arms for the ladder's top rung: content-correct scoring, sparsity pinned to the healthy 72^2 value, blur-isolation at fixed grid, and the intermediate grids.

A design principle throughout: score against real ground truth. As Section 6.4.3 shows, self-consistency (similarity of perturbed output to stock output) actively misleads — a finding that retroactively justifies the expense of GT-referenced scoring for every headline claim.

6.4 Results

6.4.1 Absolute-Position Offsets Are Free

H0 is rejected on every axis. A temporal shift of $+996$ latent positions — $\approx 50\times$ beyond the trained window — costs $+0.001$ dB against ground truth; spatial offsets of 8 and 24 latent rows cost ± 0.003 dB; and offsets remain free at every stretch level tested (no interaction). Translation invariance holds empirically on all three RoPE axes, as the relative-position algebra of RoPE predicts but distilled, finite-precision practice did not guarantee. Two engineering consequences follow: the streaming design's unbounded index growth is harmless per se, and the stock 4,089-frame single-pass ceiling is an implementation artefact of the fixed table — with the extended table, a 5,009-frame single pass ran to completion at 11.6 GiB VRAM.

6.4.2 Relative-Distance Geometry Is What Bites

Table 6.1 sweeps position-label stretch per axis with content and compute fixed. Dilation costs monotonically on the temporal axis (-0.12 dB at s=1.25 through -0.95 at s=3); strong compression costs on every axis (s=0.5: -0.66 to -1.00 dB); and mild compression — s=0.75, the position-interpolation zone — is free on every axis (+0.01/+0.02/-0.03 dB). A temporal caveat calibrates the small $s \leq 2$ costs: under streaming, FlashVSR's attention span is $\sim$8 latents, so relative distances remain inside the trained 21-latent range until $s \approx 2.6$ — those costs are within-window geometry effects, and s=3 is the first genuinely out-of-window point.

At extreme geometry distortion the axes separate sharply: space is $\sim$2.5$\times$ more sensitive than time (-2.3/-2.6 dB vs -0.95 at s=3). The continuous-position builder that underlies these numbers was validated by its identity condition (53.4 dB self-PSNR = the numerics floor) and revealed that genuine fractional-position interpolation is cheaper than integer-rounded approximations by $+0.5$–1.5 dB — integer rounding had been silently inflating compression costs.

Table 6.1 △PSNR (dB) vs stock under position-label stretch s, per RoPE axis (DOVE-UDM10 vs ground truth; content and compute fixed).

stretch s	**time (t)**	**height (h)**	**width (w)**
0.5 (compression)	-0.66	-1.00	-0.83
0.75 (PI zone)	$\mathbf{+0.01}$	$\mathbf{+0.02}$	$\mathbf{-0.03}$
1.25	-0.12	-0.05	-0.02
1.5	-0.22	-0.07	-0.03
2.0	-0.25	-0.15	-0.07
3.0 (extreme)	-0.95	$\mathbf{-2.55}$	$\mathbf{-2.30}$

6.4.3 Self-Consistency Does Not Predict Quality

Conditions $s{=}0.75$ and $s{=}1.5$ produce identical output change relative to stock (self-PSNR 31.8 for both) yet have opposite ground-truth verdicts ($+0.01$ vs -0.22 dB). How much a perturbation changes the output says nothing about whether the change damages quality. This is a methodological finding for the field: position-perturbation studies that score only against the unperturbed model's output — the cheaper and common protocol — cannot distinguish benign from harmful interventions.

6.4.4 Long-Video Drift Is Not Positional

The D″ causal check runs three arms on the full-length real-SR videos: stock segmented inference, a true single pass (positions reaching latent 1,252 on the extended table), and a magnitude-bounded arm (positions cycled mod 336). If positional magnitude drove the long-video semantic drift measured in Chapter 5, these arms would separate. They are **indistinguishable — under 1% relative spread on every video**. FlashVSR's weakest benchmark cell belongs to its streaming KV-cache/generation mechanism, not to position encoding. The check also certifies retroactively that the Chapter 5 benchmark row was fair: segmented processing did not disadvantage FlashVSR.

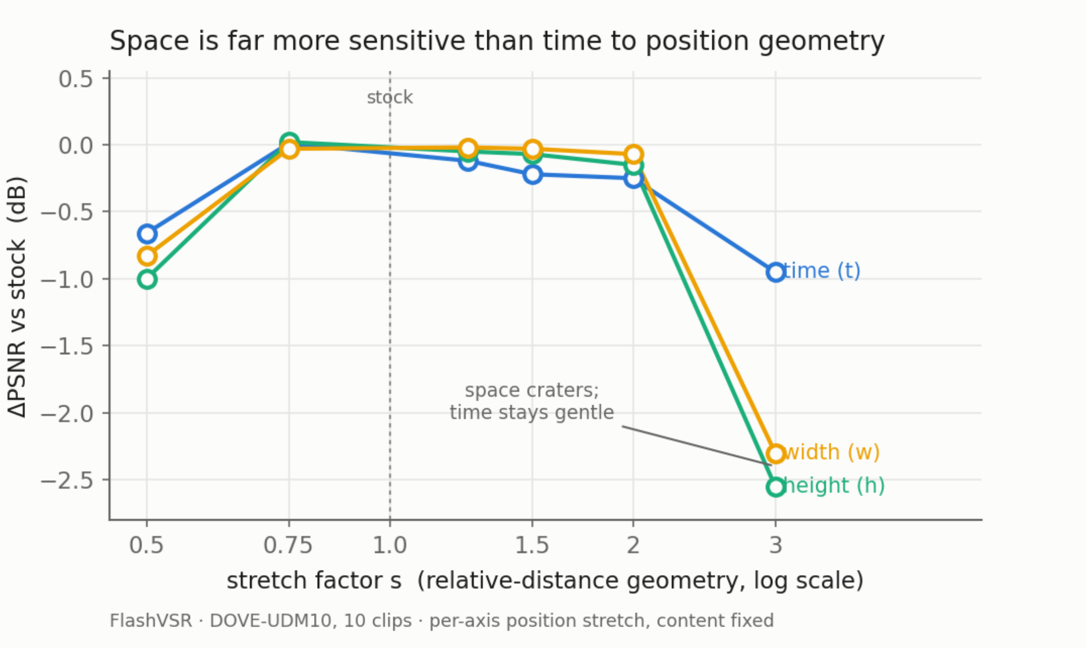

Figure 6.1 Per-axis sensitivity to position-label stretch. The PI zone (s=0.75) is free on every axis; dilation costs grow monotonically, with the spatial axes $\sim$2.5$\times$ more sensitive than time at extreme distortion.

6.4.5 Extreme Retrieval Distances

Two questions posed by the research group — "test extreme stretches" and "what does it cost when a late frame retrieves an early one at its true large relative distance, rather than a re-based small one?" — reduce to the same experiment. FlashVSR already uses true absolute indices (it never re-bases); what keeps realised relative distances small is the $\sim$8-latent KV cache, which evicts old keys long before a distance-1,000 attention edge could form. (The re-based, window-local alternative is exactly SeedVR2's design — the planned round-2 contrast model.) Holding content fixed and inflating only the position label isolates what that latent danger would cost (Table 6.2).

Three readings. First, **the loss is bounded and saturates — no collapse**: even at 50–100$\times$ the trained window, quality sits at -1.3 to -1.6 dB, barely worse than s=3's -0.95; temporal extrapolation degrades gracefully, the opposite of the spatial axes' steep -2.3/-2.6 dB at s=3 alone. Second, **the curve is non-monotonic**: s=20 recovers to -0.31 dB between ~-1 dB neighbours, because rotary phases are periodic and certain large stretches accidentally re-align fast frequencies near trained-like configurations — where the phases land, not how far they

Table 6.2 Temporal extrapolation at extreme effective distances (DOVE-UDM10 vs ground truth, continuous positions; the stock table is bypassed by the continuous builder).

stretch s	eff. distance	vs window ($\sim$20)	ΔPSNR
2	$\sim$16	window edge	-0.25
5	$\sim$40	$2\times$	-1.34
10	$\sim$80	$4\times$	-1.03
20	$\sim$160	$8\times$	-0.31
50	$\sim$400	$20\times$	-1.57
125	$\sim$1000	$50\times$ ("frame 0 from frame 1000")	-1.33
250	$\sim$2000	$100\times$	-1.56

moved, sets the damage. Third, the group's answer: the true-index design costs a bounded $\sim$1.3 dB at distance 1,000 — and FlashVSR never pays even that, because its short cache prevents the far edge from forming. Absolute index plus short cache is safe by construction, which is also why long-video drift is not positional.

6.4.6 Real Resolution Extrapolation

The label sweeps above manipulate positions at fixed content; the resolution ladder extrapolates for real, growing the latent grid from the trained extent (48^2, output 720^2) to double it (96^2, output 1536^2) on YouHQ40 (Table 6.3).

Table 6.3 YouHQ40 resolution ladder: PSNR vs ground truth per latent grid, stock positions vs spatial-PI.

output	latent grid	vs trained	stock	spatial-PI
720^2	48^2	$1\times$ (trained)	24.70	$\equiv$ stock
1152^2	72^2	$1.5\times$	24.55	24.00
1280^2	80^2	$1.67\times$	24.66	—
1408^2	88^2	$1.83\times$	24.71	—
1536^2	96^2	$2\times$	**24.78**	—

The ladder is **flat**: real resolution extrapolation is quality-free to at least $2\times$ the trained

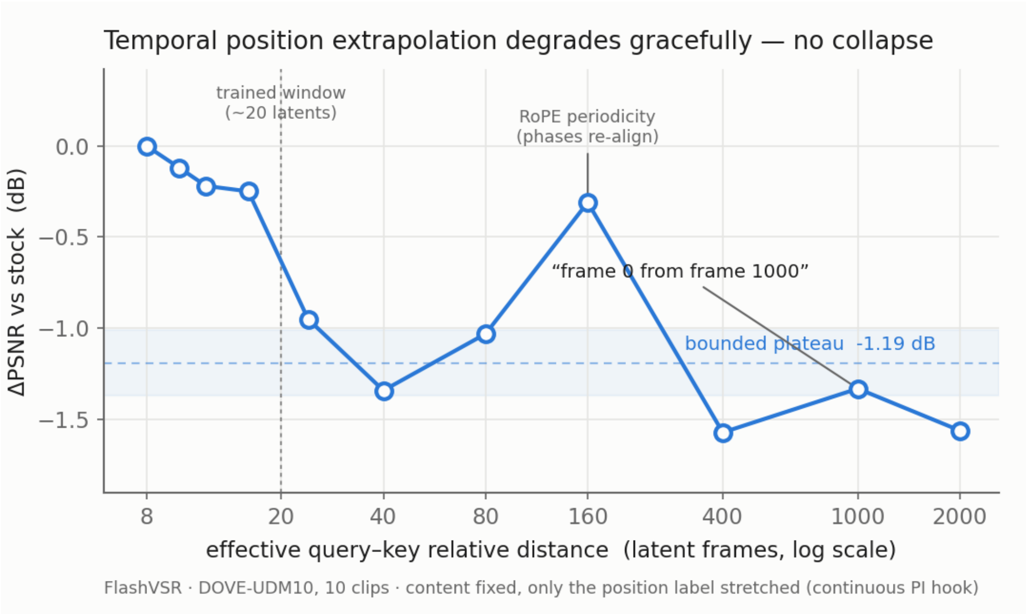

Figure 6.2 Temporal retrieval-distance dose-response: bounded, graceful, and non-monotonic. Even 100$\times$ the trained window plateaus at ~ -1.5 dB.

spatial extent with stock positions — adaptive attention sparsity included (pinning the sparsity ratio to the healthy 72^2 value changes the top rung by only -0.03 dB). The only positional effect on the ladder is that **spatial-PI at** $1.5\times$ **hurts** (-0.56 dB versus simply extrapolating): compressing positions by factor 0.67 lands in the compression-cost zone of Table 6.1. Spatially, compression — not extrapolation — is the risky direction.

A methods note, recorded honestly. An early scoring pass reported an -11 dB "collapse" at the top rung. Decomposition arms cleared every model-side suspect (sparsity pinning $\Delta -0.03$; intermediate grids flat; no blur penalty from the upsampled input), tracing the number to a scoring-geometry artefact: the padded output frame (content occupying 93.75%) had been re-sized against ground truth at mismatched magnification. This is the second instance in this thesis where a scoring-path subtlety flipped a conclusion (cf. Section 6.4.3); both were caught because raw frames and re-scorable outputs were retained — a reproducibility practice worth stating as such.

6.4.7 Why Position Interpolation Works — and Where It Stops

A fair objection to the PI results: fractional positions produce rotary phases the model never saw — are those not out-of-distribution too? The resolution[27] is the distinction between interpolation within the trained range and extrapolation beyond it. Attention consumes relative phase $(p - q)\,\theta_k$; training taught the model this function at roughly 20 discrete distances per frequency. PI places new inputs between trained points, where a learned smooth function is well-behaved; dilation places them beyond the trained interval, where the same function is unconstrained.

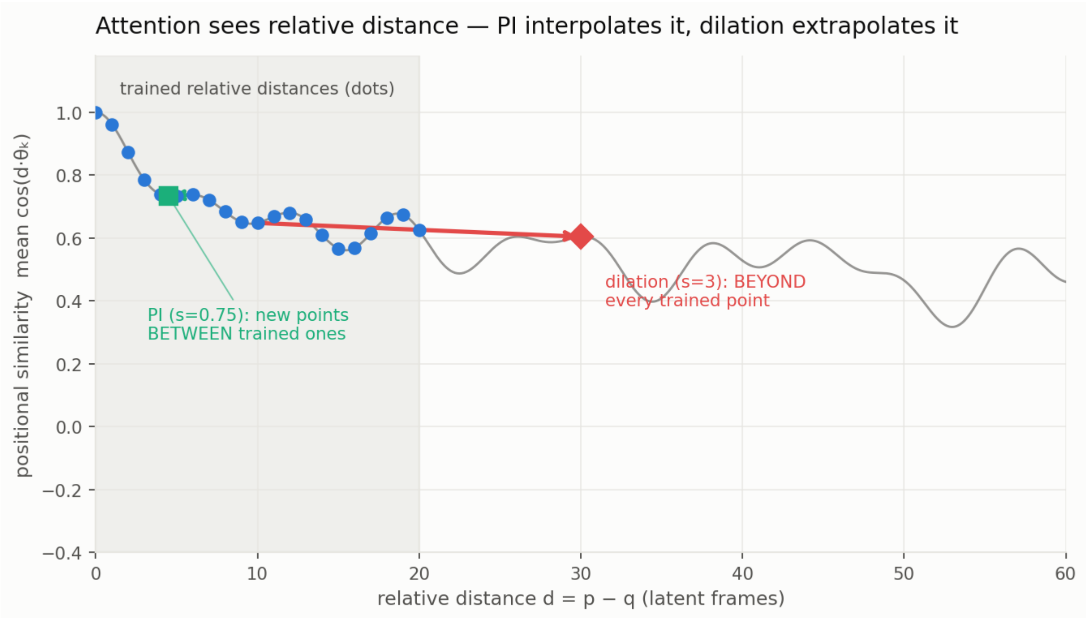

Figure 6.3 The learned relative-phase kernel: PI queries it between trained points (interpolation); dilation queries it beyond them (extrapolation).

The measurements above are this theory's per-axis fingerprint — apparently the first in a video diffusion model — sharpened by a symmetry argument: the rotation-change magnitude of a stretch depends only on $|s - 1| \cdot d$, which is symmetric between compression and dilation, yet measured quality is strongly asymmetric (s=0.75 free everywhere; extrapolation up to -2.6 dB). Where phases land decides damage, not how far they moved. The theory also predicts its own boundary: at s=0.5, neighbouring tokens sit at relative distance 0.5 — below the minimum nonzero trained distance (1.0), inside the qualitatively different self-vs-neighbour transition — which is why the zero-shot PI-free zone is narrow ($\sim$0.75) and why the LLM literature pairs aggressive PI with fine-tuning.

6.5 Implications

Position encoding is exonerated. Offsets are free on every axis at any magnitude tested; realistic geometry costs are small; the two failure modes that motivated the study — long-video drift and high-resolution degradation — are both demonstrably non-positional. Long-video improvement effort should target the streaming KV-cache/generation mechanism; the resolution path is already free to $2\times$.

Guidance for window extension. Temporal extension to $\sim 1.33\times$ ($21 \rightarrow 28$ latents) with continuous-PI positions is predicted RoPE-loss-free; beyond that, temporal geometry costs grow but stay bounded and graceful — even $100\times$ the window plateaus at ~ -1.5 dB, so RoPE will not be the wall a window extension hits. The validated instrument transfers directly: comparing stock vs PI positions in extended-window runs isolates RoPE's causal share.

Do not transplant LLM context-extension recipes blindly. PI helps only in its narrow zero-shot zone and actively hurts at $\leq 1.5\times$ spatial extension, where stock extrapolation is cheaper. Per-axis measurement, not one-dimensional intuition, is the transferable practice.

6.6 Limitations

All ground-truth evidence is temporally within-window (29–31-frame GT clips); deep-length ($>$81-frame) quality-vs-position curves lack ground truth, as re-acquisition of long HR sources is blocked platform-side — the path forward is lab-provided long HR footage, for which conversion tooling is ready. The study covers a single model (FlashVSR 1.3B distilled); SeedVR2, with window-local positions, is the designated contrast for a second round. The spatial trained extent is inferred from the distillation resolution rather than training logs.

Chapter 7. Discussion and Conclusion

7.1 Significance

Long-video super-resolution evaluation now has a no-reference, multi-time-scale, reliability-gated instrument whose every design decision traces to a characterised failure mode — and whose failure modes are, in turn, characterised rather than hidden. That closed loop, more than any single number, is the methodological contribution: metrics for generative video are themselves measurement instruments that require calibration, control batteries, and reproduction gates, and this thesis demonstrates what that discipline looks like in practice.

The failure-mode characterisations stand on their own. The smoother-output bias — three independent learned representations all rewarding the blurrier method, to the point that a state-of-the-art consistency benchmark ranks a degraded input above its restorations — reframes a whole family of published temporal metrics as partial spatial-smoothness measurements. The temporal-scale crossover shows that the literature convention of reporting tOF at k=1 inverts method rankings against long-range behaviour. The convergence-rewarding pathology shows that stability-style consistency scores have a structural failure class on corruptions that make videos more static. Each of these is actionable by any group building or using video quality metrics, independent of LR-VCC.

The RoPE study contributes both verdicts and method. The verdicts — absolute position magnitude is free at any tested scale on every axis, relative-distance geometry is what bites (gracefully in time, steeply in space), long-video drift and high-resolution degradation are demonstrably non-positional, and real resolution extrapolation is free to twice the trained extent — redirect engineering effort for streaming diffusion VSR away from position encoding. The method — a bit-exact position-injection instrument, ground-truth-referenced scoring after demonstrating that self-consistency misleads, and per-axis decomposition of extrapolation into magnitude and geometry — transfers to any RoPE-based video model.

7.2 Limitations

The benchmark's scale supports mechanism-level claims, not community-leaderboard claims: five base videos, twelve artefact families, and a handful of SR methods. Content with frequent hard scene cuts defeats the fixed opening anchor of the drift sub-metrics; pure horizontal mirroring is invisible to all seven sub-metrics; the identity families remain the weakest matrix rows, with the dispersion gate parked for want of calibration data. The FlashVSR-versus-MGLD-VSR comparison depends on the identity sub-metric's weight and softmax sharpness, and is reported as such — only MGLD $>$ UAV is unconditional. On the RoPE side, all ground-truth evidence is temporally within-window; deep-length quality-versus-position curves await long HR footage; and the study covers one model.

7.3 Future Work

Six directions follow directly from the documented limitations. Adaptive anchoring: re-establish the D'/D'' anchor at detected scene cuts, converting the cartoon-content failure into a content-aware design. Mirror sensitivity: a pixel-correlation or pose-based sub-metric to close the flip_horizontal blind spot. Identity-gate calibration: an 8-plus-base single-face corpus to unpark the dispersion gate and repair the slow-fast pooling pathology. Benchmark scale-up: more bases, more methods, and human-judgement calibration toward a public leaderboard. SSM temporal backbones: the long-context state-space direction[30] that motivated this work initially — with LR-VCC now available as a consistency-aware evaluation signal, or even a training signal. RoPE round two: the SeedVR2 window-local-position contrast, the extended-window runs with the handed-over instrument, and deep-length ground-truth curves once lab HR footage lands.

7.4 Conclusion

This thesis asked whether long-form video super-resolution can be evaluated honestly with the instruments the field currently uses, showed empirically that it cannot, and built the missing instrument. LR-VCC composes seven reliability-gated sub-metrics into a no-reference consistency score that catches the long-range failure modes existing metrics are blind to, survives the content regimes that flip existing metrics against human judgement, and exposes its own remaining blind spots by design. Applied to real methods, it produced a ranking per-frame metrics cannot see and a diagnosis — streaming long-range drift — that its own follow-up study then localised causally: position encoding exonerated, generation mechanics implicated. The evaluation gap for long videos is real, structural, and closable; this thesis closes the characterised part of it and maps the road for the rest.

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